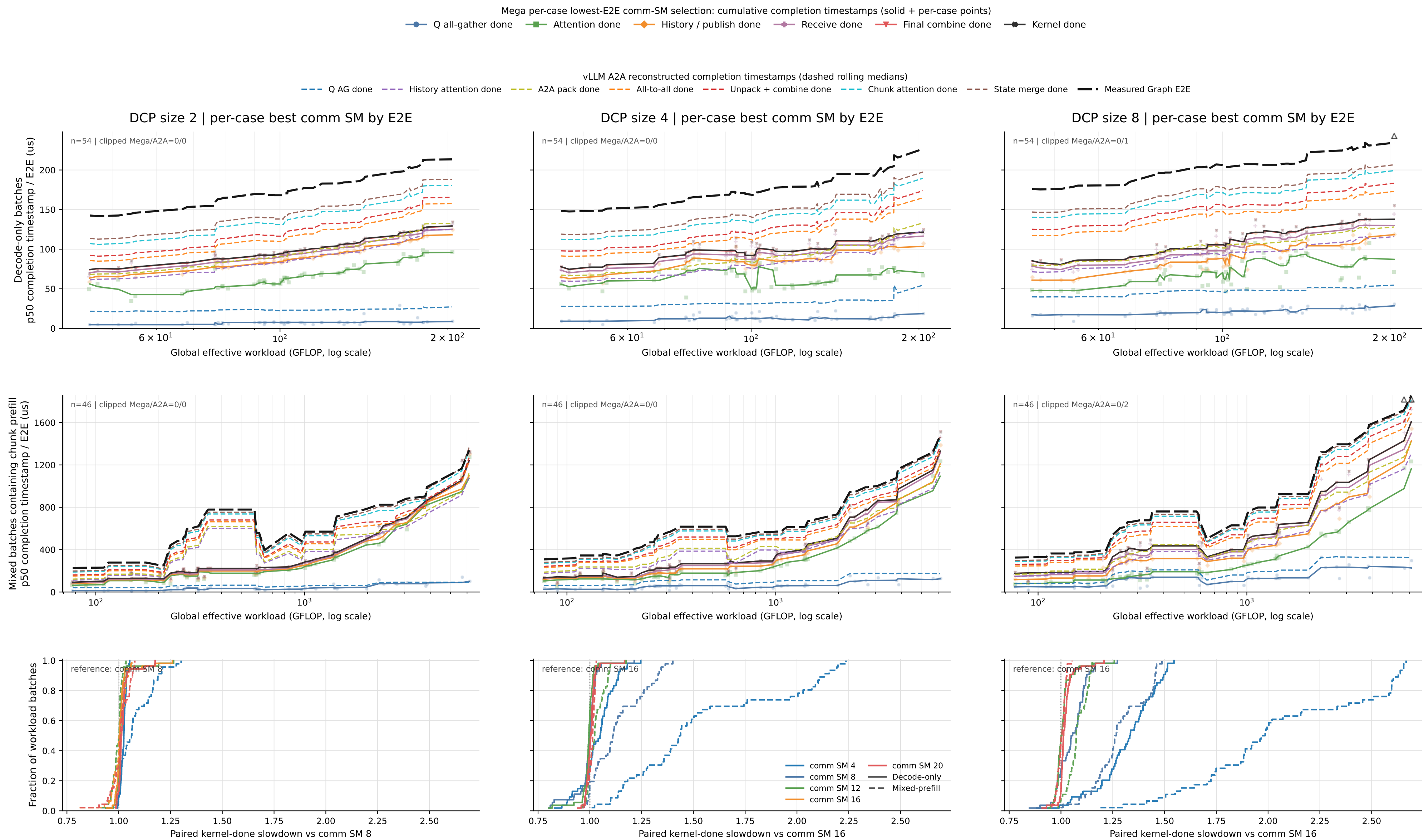


Mega DCP and vLLM A2A CUDA Graph Phase Latencies

Arrival scale 4 | 100 paired workload batches (54 decode-only, 46 mixed-prefill) | 60 measured iterations per case | Mega uses each case's lowest-E2E comm SM | Mega and vLLM A2A Graph paired by case | source: 20260812-125820-arrival4-phases-graph



Decode: Start -> attention	61.5	65.2	66.5	68.0	68.7
Decode: Attention -> publish	28.6	25.9	24.4	24.3	24.2
Decode: Publish -> receive	11.4	8.8	8.8	8.7	8.7
Decode: Receive -> final	3.9	3.9	3.8	3.9	4.0
Decode: Start -> kernel done	99.0	96.9	98.1	97.4	99.1
Mixed: Start -> attention	197.0	197.2	192.2	191.1	192.4
Mixed: Attention -> publish	18.3	17.3	19.1	19.3	21.1
Mixed: Publish -> receive	25.3	13.8	13.7	12.1	11.1
Mixed: Receive -> final	10.8	10.4	11.2	11.4	11.5
Mixed: Start -> kernel done	257.1	236.2	226.9	223.1	223.3
	4	8*	12	16	20

Communication SMs (* aggregate best)

63.6	65.5	68.2	72.9	73.9
21.2	16.6	13.7	14.0	13.7
16.6	10.8	9.9	8.9	8.4
4.7	4.8	4.8	4.9	4.9
107.5	101.8	101.0	102.9	102.5
188.5	187.2	182.3	178.3	183.6
37.6	26.4	27.5	29.4	31.3
247.4	103.4	55.0	37.5	24.9
14.6	14.7	15.0	15.8	21.3
503.4	323.2	280.0	267.3	262.8
4	8	12	16*	20

Communication SMs (* aggregate best)

75.8	69.2	71.1	72.2	75.3
14.0	16.8	17.0	18.4	18.6
53.7	21.6	12.4	11.9	11.7
7.0	7.2	7.0	7.0	6.8
147.8	120.0	115.3	113.8	117.7
310.5	199.8	178.9	180.5	185.1
111.6	122.3	129.8	132.1	141.5
585.6	244.3	140.2	87.0	50.2
25.1	25.5	25.7	27.8	53.2
1032.9	608.1	474.5	416.5	415.4
4	8	12	16*	20

Communication SMs (* aggregate best)

Median per-case p50 tail duration (us)